Answer Key - Physics - Undergraduate Level Assessment 24 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** D
   Options: A) square; B) hexagon; C) cube; D) tetrahedron
   Note: Correct option: D - tetrahedron

2. **Answer:** D
   Options: A) Bosons have symmetric wave functions and obey the Pauli exclusion principle.; B) Bosons have antisymmetric wave functions and do not obey the Pauli exclusion principle.; C) Fermions have symmetric wave functions and obey the Pauli exclusion principle.; D) Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.
   Note: Correct option: D - Fermions have antisymmetric wave functions and obey the Pauli exclusion principle.

3. **Answer:** C
   Options: A) There are no changes in the internal energy of the system.; B) The temperature of the system remains constant during the process.; C) The entropy of the system and its environment remains unchanged.; D) The entropy of the system and its environment must increase.
   Note: Correct option: C - The entropy of the system and its environment remains unchanged.

4. **Answer:** A
   Options: A) rate of change of the electric flux through S; B) electric flux through S; C) time integral of the magnetic flux through S; D) rate of change of the magnetic flux through S
   Note: Correct option: A - rate of change of the electric flux through S

5. **Answer:** D
   Options: A) magnetic flux through S; B) rate of change of the magnetic flux through S; C) time integral of the magnetic flux through S; D) rate of change of the electric flux through S
   Note: Correct option: D - rate of change of the electric flux through S

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '4'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'interactions with the ionic lattice'.

Long Form Response

11. **Answer:** 10 eV. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2 V_0/3. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key